COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-008: Network Monitoring

Document ID	TS-INF-008
Title	Network Monitoring
Executive Sponsor	Network Services, Sr. Manager of Network Operations
Issuing Authority	Office of Information Technology (OIT)
Applies to	State of Colorado agencies; defines the requirements and specifications for network monitoring within the State of Colorado.
Tier	Standard
Version	3.0
Effective Date	11/01/2022
Last Updated	12/31/2025

1. Purpose

The purpose of the network monitoring standard is to provide network administrators and engineers with valuable insights into the health and performance of the State's network. This proactive approach helps ensure optimal network operation, identify and resolve issues before they become critical, and plan for future network needs.

2. Scope

This technical standard defines the requirements and specifications for network monitoring within the State of Colorado agencies. It focuses on the information collection function of network management systems that monitor and manage the State of Colorado networks, and outlines: Data Collection; Network Management Applications interoperability requirements; Device State Monitoring; and Performance and Alert Monitoring.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Network Services - Sr. Manager of Network Operations

- Owner of Standard; engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number; presents to the ARB for approval and adoption.

Network Services - Network Operations Staff

- Keeps the application updated and verifies the system has sufficient resources.
- Maintains inventory on the system; builds and provides reports; handles alerting and remediation.

Server Team

- Keeps the server operating system updated and free from vulnerabilities.
- Performs weekly backups; assists in restoring the system.

Database Team

- Maintains the SQL database for the network monitoring application.
- Provides the correct resources on the database server; maintains and manages database backups.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Monitoring Requirements

- All network devices capable of being monitored must be monitored (routers, switches, wireless controllers, access points, load balancers, and firewalls).
- The monitoring solution must collect different metrics at configurable intervals to aid in scaling.
- Must support: SNMP Version 3 (requests and traps); Syslog; NetFlow collection (desirable); and custom SNMP OIDs.

Alerting Requirements

- Must generate alerts based on configurable criteria with varying severity levels (minor, major, critical).
- Must support criteria around: time of day; day of week; device name; device grouping; location; type of device; device component; type of metric; metric value or threshold; severity; compound conditions; and hold-downs/suppression.
- Must support forwarding alerts to external services (email, SMS, Google Hangout).

Metric Requirements

- Must collect: availability (up/down); delay; interface utilization; interface errors; CPU utilization; and hardware sensors.
- Availability, delay, interface utilization, interface errors, and CPU utilization must be polled at an interval not exceeding 10 minutes, ideally every 5 minutes.
- Wireless metrics: number of wireless clients per AP; total wireless users; wireless interference; SNR; and wireless coverage with maps.

Data Retention Requirements

- All metrics must be stored and provide historical information for at least one year.
- Rollups at configurable periods for reducing dataset size are acceptable; recommended periods: 30 days, 90 days, and 1 year.
- The application server and database must be backed up once a week, with a minimum of 4 weeks of backups stored.

Reporting Requirements

- Reporting capability must be provided as part of the solution, supporting customized reporting.
- Reports must be available for all required metrics for both devices and device groups.
- Must support scheduling, emailing, and recurring reports.
- All reports must support HTML, PDF, Excel, and CSV formats.
- Must support sending logs to an enterprise SIEM (Splunk, presently).

System Location Requirements

- The network monitoring application must be on a separate network from the systems it is monitoring.
- Must be deployed behind a robust firewall with strict access controls, configured to allow only the minimum necessary ports for operation.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the "Owner of Standard" listed under Roles and Responsibilities.

Requests may be acquired by completing the "EXCEPTION REQUEST to Technical Standard" form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

- Techopedia: Network Performance Monitoring
- SolarWinds
- Washington University: Network Monitoring resources

7. Definitions

Network Monitoring: The use of a system that constantly monitors a network for slow or failing components and notifies the network administrator of outages or troubles.

NetFlow: Network protocol for collecting and analyzing network traffic data, providing insight into traffic patterns, bandwidth usage, and potential security threats.

SNMP: Simple Network Management Protocol — used for monitoring and managing network devices, collecting performance data, remote configuration, and receiving alerts.

Network availability: Overall uptime of a network.

Network delay: How long it takes for a bit of data to travel across the network from one node/endpoint to another.

Interface utilization: Data transfer rate, usually expressed in bits per second (bps).